InsureVista 360°: Project Write-up

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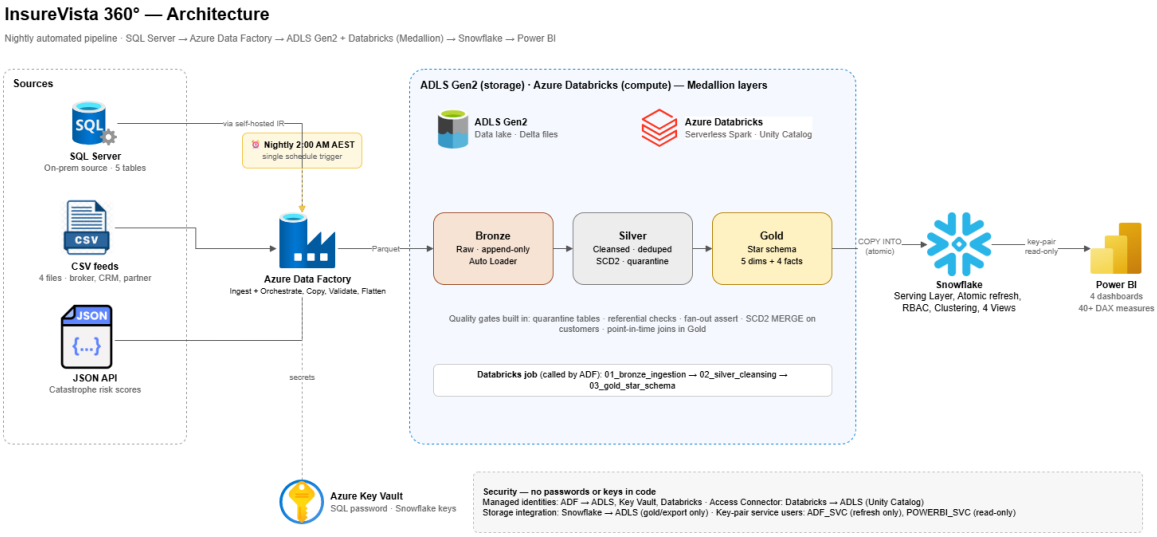
Summary

InsureVista 360° is an end-to-end insurance analytics platform, built for a simulated Hollard Insurance Australia, that runs nightly from an on-premises SQL Server to four Power BI dashboards with no manual steps.

It moves 513,781 source rows (440,320 from five SQL Server tables, plus four CSV feeds and one JSON API feed) through a Medallion lakehouse (Bronze, Silver, Gold) into a nine-table star schema, served from Snowflake to Power BI.

- **Stack:** SQL Server, Azure Data Factory, ADLS Gen2, Azure Databricks (Unity Catalog, serverless), Snowflake, Power BI, Azure Key Vault.
- **Automation:** one ADF schedule at 2:00 AM AEST runs five chained steps; any failure stops the chain and rolls back the Snowflake load.
- **Proof:** every layer reconciles to the cent. Incurred claims total $470,534,897.33 and premium received $776,976,280.62 from SQL Server through to Power BI.
- **Security:** no passwords or keys in code. Azure services use managed identities; Snowflake service users use RSA key pairs stored in Key Vault.
- **Outcome:** four dashboards (Executive Summary, Claims Intelligence, Customer 360°, Broker & Partner Performance) on a 60.6% overall loss ratio after data calibration.

Architecture

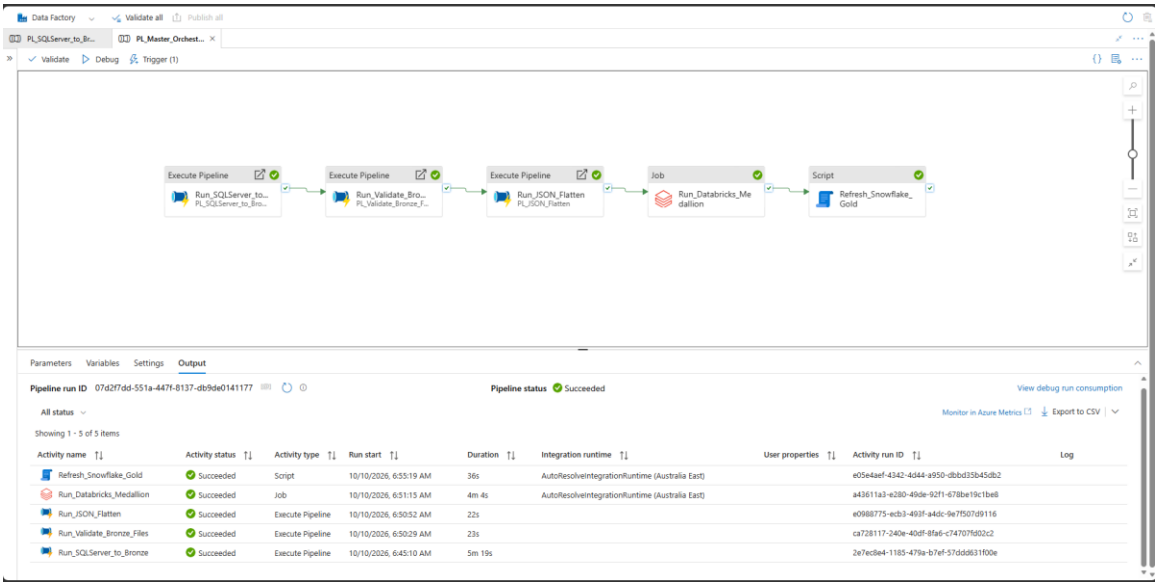


Data flows left to right. Azure Data Factory lands the three sources in Bronze on a nightly 2:00 AM AEST schedule; a Databricks job refines Bronze to Silver to Gold in ADLS Gen2; Snowflake loads Gold atomically; Power BI reads Snowflake through a read-only key-pair user. The security band shows how every hop authenticates without a password or key in code.

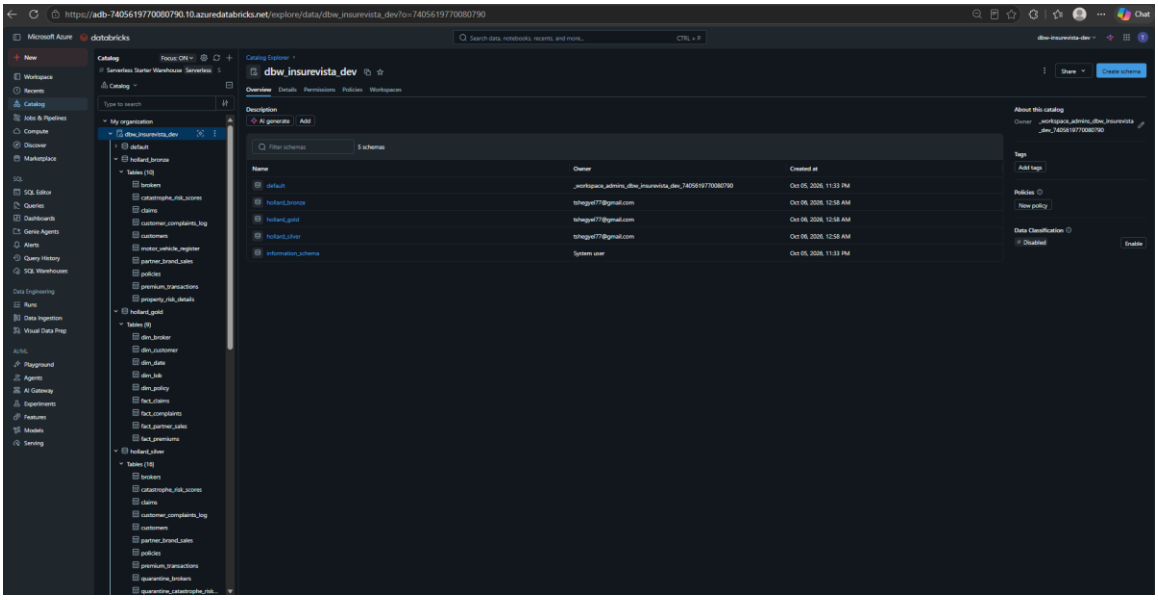
Layer by layer

Each tool has one job, and no two tools overlap.

Layer	Tool	What it does	Key detail
Source	SQL Server 2025 (local)	Simulates the on-prem policy system: Customers, Brokers, Policies, Claims, Premium_Transactions	Foreign keys enforced at source, so zero orphans arrived downstream
Ingest	Azure Data Factory + self-hosted IR	Copies 5 SQL tables to Parquet, validates 5 file feeds, converts JSON to Parquet	Parameterised ForEach over one dataset pair; retries (2 × 60 s) absorb SHIR blips
Bronze	Databricks Auto Loader	Appends every file once into 10 Delta tables with audit columns	Checkpointed: a re-run loads +0 rows; `_batch_id` carries the ADF run ID
Silver	Databricks (PySpark)	Deduplicates, casts, normalises, enforces referential integrity, quarantines bad rows	SCD Type 2 on customers via MERGE; hard rules quarantine, soft rules warn
Gold	Databricks (PySpark)	Builds the star schema: 5 dimensions, 4 facts	Hash surrogate keys, Unknown (-1) members, point-in-time customer joins
Serve	Snowflake	Loads Gold atomically, clusters facts, enforces roles, exposes 4 semantic views	`SP_REFRESH_GOLD` rolls back all 9 tables if any load fails
Present	Power BI Desktop	Star-schema model, 40+ DAX measures, 4 dashboards	Import mode; key-pair service user with read-only role
Orchestrate	ADF schedule	Chains all five steps nightly at 2:00 AM AEST	One trigger; Sydney-dated Bronze folders, DST-aware



One orchestrator run end to end: SQL Server copy, Bronze validation, JSON flatten, the Databricks Medallion job and the Snowflake refresh, all succeeded in about 11 minutes.



The Medallion layers in Unity Catalog: 10 Bronze tables, 16 Silver tables (including quarantine), 9 Gold star-schema tables.



The Power BI star schema: single-direction relationships, inactive role-playing date links, and a dedicated measures table.

Design decisions and deviations from the spec

The original design document was followed in structure, but ten choices were changed after inspecting the data or testing the result.

Area	Spec said	Built instead	Why
JSON feed	Mapping Data Flow to flatten nested JSON	Copy activity, JSON to Parquet	The feed is a flat array of objects; no Spark cluster needed, 14 s per run
SQL watermark	Incremental load on `modified_date`	Full daily snapshot, deduplicated in Silver	Source tables have no change column; Auto Loader keeps Bronze append-only
Bronze write	`mode("overwrite")`	Auto Loader with checkpoints, append	Overwrite would duplicate or lose history; Bronze now

Area	Spec said	Built instead	Why
			replays any day
Dedup ordering	`ORDER BY _ingestion_ts DESC`	Snapshot date from the file path, then ingestion time	Two daily loads shared one ingestion timestamp, so the spec's rule picked at random
Surrogate keys	Auto-generated INT	`xxhash64` of the business key	Stable across rebuilds; row-number keys shift when a row is added
Unknown members	Not specified	Key -1 in every dimension	The spec's INNER-joined views would silently drop facts with missing keys
Gold to Snowflake	Stage over the Delta folder	Separate plain-Parquet export folder	Delta keeps old file versions; `COPY INTO` would load them all
Snowflake auth	SAS token in the stage	Storage integration (managed identity)	No secret in SQL, no expiry
Semantic views	Shown in the spec	Rewritten (see Bugs found)	Fan-out overstated claims 4×; `is_current` filter hid history
Power BI mode	DirectQuery for facts	Import	350,000 rows fit in memory; no warehouse wake-up per click

Bugs found and fixed

Nine defects were caught by reconciling numbers rather than trusting output that looked plausible.

#	Symptom	Root cause	Fix	Evidence
1	Spec's LOB and broker views reported wrong totals	Fact-to-fact join through a shared dimension (chasm trap): each premium × each claim per policy	Aggregate each fact separately, then join	Spec pattern: $16.69B claims vs $4.15B true (4.0×); premium 2.5×

#	Symptom	Root cause	Fix	Evidence
2	Pre-move claims would vanish from dashboards	Spec view filtered `is_current = 1`, but point-in-time keys point to historical versions	Removed the filter	Customer HL-CUS-00001's 2024 claim stays linked to Brisbane CBD
3	Customer lifetime value overstated	`SUM()` over an SCD2 dimension counts every version	Measure filtered to current rows	Naive sum $42,140.86 too high after one customer moved
4	Every NatCat bar showed the same value	`CALCULATE` boolean filter replaced the visual's own filter on that column	`KEEPFILTERS()` on nine measures	Per-event costs now $4.48M to $4.98M
5	"Top 10 claim types" ranked alphabetically	Top N filter defaulted to *First CLAIM_TYPE*	Rank by `[Incurred Claims]`	Commercial types moved to the top at $48–51M each
6	Read-only role test was not isolated	Snowflake secondary roles let ACCOUNTADMIN authorise in the background	`USE SECONDARY ROLES NONE` in tests; service users hold one role only	Re-test failed with "insufficient privileges" as intended
7	Child pipeline rejected its array parameter	Parameter arrived as a JSON-looking string via Execute Pipeline	`@json(string(...))` accepts either form	Orchestrator ran all three children
8	Bronze reads returned 2× rows	Two date folders read recursively	Auto Loader with checkpoints; Silver dedup	640 broker rows collapsed to 320
9	Nightly SQL copy failed once with no config change	Self-hosted IR reconnecting after an update	Copy retries: 2 × 60 s	Subsequent runs succeeded

The common thread: each bug produced a believable chart. Every one was found by checking a total against its source.

Data findings and caveats

Profiling showed the synthetic source was structurally clean but statistically unrealistic, so Silver rules were kept as guardrails and the unrealistic parts were calibrated or disclosed.

Structurally clean. Zero null keys, zero orphan foreign keys in every relationship checked, and no spelling variants in any status column. The spec's cleansing rules (status normalisation, quarantine) therefore catch zero rows today; they are kept as protection against future feed drift.

Real issues found and handled:

- **Placeholder campaign codes.** 9,385 partner sales (37.5%) carried `'NONE'`, which would have shown as the largest campaign. Converted to NULL in Silver.
- **Events before the customer existed.** 6,122 claims (4.7%) and 10,511 premium payments (4.8%) pre-date the customer's `customer_since` date. The point-in-time join falls back to the customer's earliest version, so no rows are lost.
- **Unrealistic loss ratio.** Claims were 534% of premium, against roughly 55–75% for Australian general insurers. Claim amounts were calibrated in SQL Server with a separate factor per line of business, giving a 60.6% overall loss ratio. The original table is kept as `dbo.Claims_backup_20261008`.

Disclosed rather than fixed:

- **Life claim frequency.** 6,274 Life claims on 2,692 policies (2.3 per policy). Scaling amounts cannot fix frequency, so Life is calibrated on ratio only and its average claim ($146) is not meaningful.
- **Uniform distributions.** Settlement days (about 63), fraud rate (about 4%), complaint categories, sales channels and campaign ROI are near-identical across groups. NPS scores are uniform from 1 to 10, which mechanically yields an NPS of −40.
- **Retention.** 49.5% is a proxy (renewed versus lapsed or cancelled policies) because the source has no renewal history.

Each dashboard page carries a footnote with the relevant caveat.

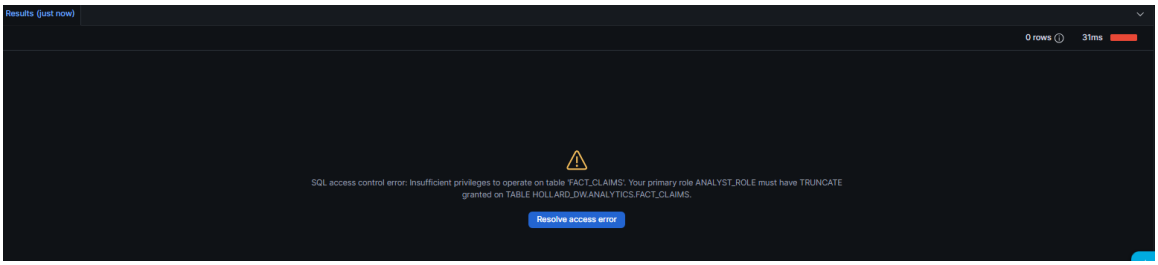
Security and governance

No password, key or token appears in any pipeline, notebook or SQL file.

Connection	Identity	Secret storage	Scope
ADF → SQL Server	SQL login	Key Vault (`sql-sa-password`)	Via self-hosted IR on the local machine
ADF → ADLS, Key Vault, Databricks	ADF system-assigned managed identity	None needed	Blob Data Contributor; Secrets User; workspace Contributor
Databricks → ADLS	Access Connector managed identity	None needed	Unity Catalog storage credential + 3 external locations

Connection	Identity	Secret storage	Scope
Snowflake → ADLS	Storage integration (Entra app)	None needed	Blob Data Reader on the `gold` container; allowed path `gold/export/` only
ADF → Snowflake	`ADF_SVC`, TYPE = SERVICE	RSA key + passphrase in Key Vault	`LOADER_ROLE`: may call `SP_REFRESH_GOLD`, cannot read any table
Power BI → Snowflake	`POWERBI_SVC`, TYPE = SERVICE	Encrypted RSA key on the local machine	`ANALYST_ROLE`: read-only

Least privilege, tested. Each role was proven by a deliberate failure: the analyst role cannot TRUNCATE; the loader role can refresh all nine tables through the owner's-rights procedure yet gets "object does not exist or not authorized" on a direct SELECT.



Least privilege proven: ANALYST_ROLE is refused a TRUNCATE on FACT_CLAIMS (tested with secondary roles off).

Other controls:

- Data stays in Azure Australia East and Snowflake Azure Australia East (onshore).
- Contact details (email, mobile) stay in Silver and are excluded from Gold and Power BI.
- Unity Catalog provides lineage across Bronze, Silver and Gold.
- Silver writes rejected rows to eight `quarantine_*` tables with a reason per row.
- A Snowflake resource monitor suspends the warehouse at 50 credits a month.
- Each service user has its own key pair, so revoking one never breaks another.

Regulatory hooks: AFCA escalations (436, 5.5% of complaints) and IDR escalations are tracked monthly for ASIC RG 271 reporting.

Reconciliation and results

The same totals were proven at every layer, so a number on a dashboard can be traced to its source row.

Check	SQL Server	Silver	Gold	Snowflake	Power BI
Claims (rows)	130,000	130,000	130,000	130,000	130,000

Check	SQL Server	Silver	Gold	Snowflake	Power BI
Incurred claims ($)	470,534,897.33	470,534,897.33	470,534,897.33	470,534,897.33	470,534,897.33
Premium received ($)	776,976,280.62	776,976,280.62	776,976,280.62	776,976,280.62	776,976,280.62
Partner premium ($)	— (CSV feed)	105,364,325.51	105,364,325.51	105,364,325.51	105,364,325.51
Customers	35,000	35,001 rows (1 history)	35,002 incl. Unknown	35,002	35,000 current

Headline results (Executive Summary, all years):

- Gross written premium $777M; incurred claims $471M; loss ratio 60.6%.
- Loss ratio by line ranges from 55% (Commercial) to 85% (Health); Tasmanian Health is the hot spot at 99%.
- Collection rate 84.3%; fraud-flagged claims 4.0%; claims funnel 130K lodged to 84K approved (64.8%).
- Top claim costs are Commercial: public liability, business interruption, property damage, professional negligence ($48–51M each).
- Nine catastrophe events cost $42.2M, about 9% of incurred claims.
- Partner channels: $105M premium at a 16.5% acquisition cost ratio.

Pipeline runtime (one full debug run): SQL copy 2–10 min depending on retries, validation 1 min, JSON 1.5 min, Databricks Bronze→Silver→Gold about 4–5 min, Snowflake refresh under 1 min.

Output

Copy to clipboard


```
{
    "resultSetCount": 1,
    "recordsAffected": 0,
    "resultSets": [
        {
            "rowCount": 9,
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                {
                    "TABLE_NAME": "DIM_DATE",
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                },
                {
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                },
                {
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                    "ROWS_LOADED": 321
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                    "TABLE_NAME": "DIM_CUSTOMER",
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                {
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                    "ROWS_LOADED": 55001
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        "activityType": "PipelineActivity",
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                "meterType": "AzureIR",
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            }
        ]
    }
}
```


ADF calls SP_REFRESH_GOLD and Snowflake returns the row count for all 9 tables, matching Gold exactly, in 22 seconds.

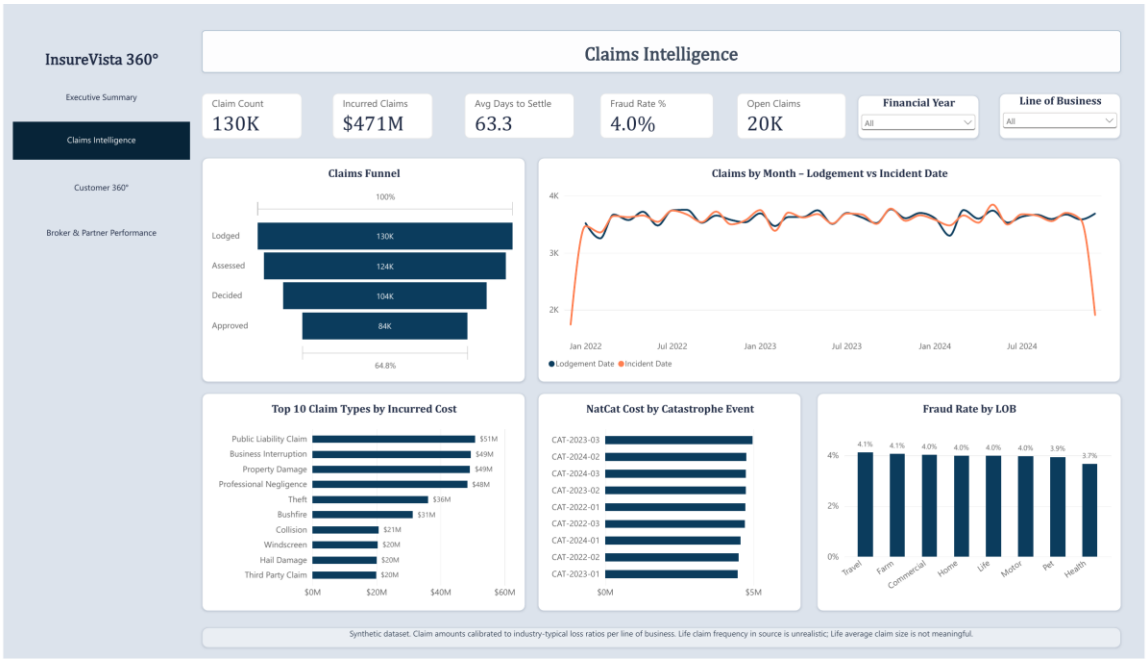
Idempotency proven: re-running Bronze loads +0 rows, re-running the customer MERGE affects 0 rows, and a fourth daily snapshot leaves every Silver and Gold total unchanged.

Dashboards

Four Power BI pages share one theme, one Financial Year and Line of Business filter, and the reconciled numbers above.



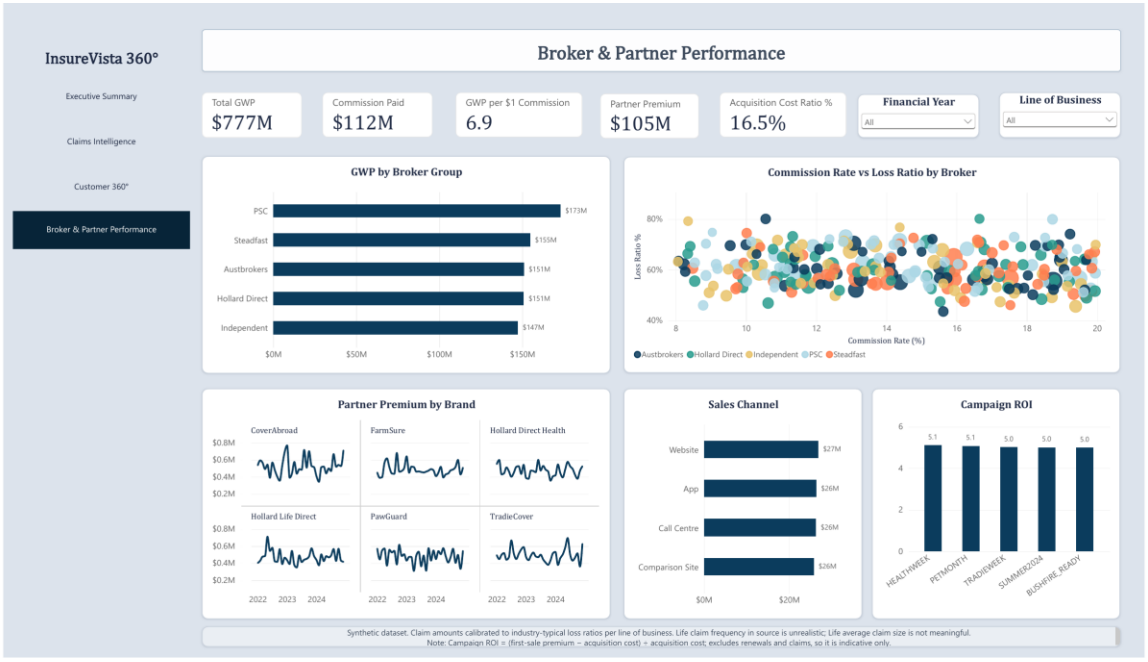
Executive Summary: $777M GWP, $471M incurred, 60.6% loss ratio, loss ratio by line of business against break-even, and a state × line heatmap.



Claims Intelligence: the claims funnel from lodged to approved, lodgement vs incident date trends, top claim types, NatCat cost by event and fraud rate by line.



Customer 360°: complaints by category, AFCA and IDR escalations, NPS distribution, retention by line and customers by risk tier.



Broker & Partner Performance: GWP by broker group, commission rate against loss ratio per broker, partner brand trends, sales channels and campaign ROI.

Limitations

The platform runs unattended, but four limitations remain and are listed with the change that would close each.

Limitation	Effect	Production fix
SQL Server runs on a laptop	The 2:00 AM run needs the machine awake and the IR connected	A server or Azure SQL source
Same-day re-runs overwrite files at the same path	Auto Loader skips them, so a changed same-day file would be missed	Timestamp in the file name, or `cloudFiles.allowOverwrites`
Power BI Desktop only	Refresh is manual after the nightly load	Publish to the Power BI Service with a scheduled refresh
No failure alert configured	Failures are seen in ADF Monitor, not pushed	Azure Monitor alert on failed runs of the master pipeline (needs `Microsoft.Insights` registered)